Actividad #1

Lee el siguiente artículo e instala Apache en Ubuntu: https://www.digitalocean.com/community/tutorials/how-to-install-linux-apache-mysql-php-lamp-stack-on-ubuntu-20-04-es

Lleva a cabo todos los pasos descritos en el artículo, incluído:

- instalación de Apache

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo apt update
[sudo: authenticate] Password:
Obj:1 http://security.ubuntu.com/ubuntu questing-security InRelease
Obj:2 http://archive.ubuntu.com/ubuntu questing InRelease
Obj:3 http://archive.ubuntu.com/ubuntu questing-updates InRelease
Obj:4 http://archive.ubuntu.com/ubuntu questing-backports InRelease
Todos los paquetes están actualizados.
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo apt install apache2
Installing:
  apache2
```

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo ufw app list
Aplicaciones disponibles:
  Apache
  Apache Full
  Apache Secure
  CUPS
  Wsdd
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo ufw allow in "Apache"
Omitiendo adición de regla ya existente
Omitiendo adición de regla ya existente (v6)
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo ufw status
Estado: activo

Hasta                      Acción      Desde
-----                      ------      -----
Apache                     ALLOW       Anywhere
Apache (v6)                ALLOW       Anywhere (v6)
```

- Instalación de MySql

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo apt install mysql-server
Installing:
  mysql-server

Installing dependencies:
  libaio1t64          libgoogle-perftools4t64  mecab-ipadic-utf8
  libcgi-fast-perl    libhtml-template-perl    mecab-utils
  libcgi-pm-perl      libmecab2                mysql-client
  libfcgi-bin         libprotobuf-lite32t64    mysql-client-core
  libfcgi-perl        libtcmalloc-minimal4t64  mysql-common
  libfcgi0t64         mecab-ipadic             mysql-server-core

Paquetes sugeridos:
  libipc-sharedcache-perl  mailx  tinyca

Summary:
  Upgrading: 0, Installing: 19, Removing: 0, Not Upgrading: 0
  Download size: 31,1 MB
  Space needed: 237 MB / 19,5 GB available
```

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo mysql_secure_installation

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

VALIDATE PASSWORD COMPONENT can be used to test passwords
and improve security. It checks the strength of password
and allows the users to set only those passwords which are
secure enough. Would you like to setup VALIDATE PASSWORD component?

Press y|Y for Yes, any other key for No: y

There are three levels of password validation policy:

LOW    Length >= 8
MEDIUM Length >= 8, numeric, mixed case, and special characters
STRONG Length >= 8, numeric, mixed case, special characters and dictionary
           file
```

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.4.10-0ubuntu0.25.10.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> exit
Bye
```

- Instalación de PHP

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo apt install php libapache2-mod-php php-mysql
[sudo: authenticate] Password:
apt: symbol lookup error: apt: undefined symbol: _ZN3APT6Solver14InternalCliWhyB5cxx11ER11pkgDepCach
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$
```

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo apt install php libapache2-mod-php php-mysql
Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios:
  cpdb-backend-cups             libboost-iostreams1.88.0  liblastlog2-2                  libsoup-2.4-1      mousetweaks
  fonts-font-awesome            libboost-locale1.88.0     liblqr-1-0                     libsoup2.4-common  python3-dateutil
  fonts-lato                    libboost-thread1.88.0     libmagickcore-7.q16-10         libtheora0         python3.13
  gir1.2-girepository-2.0       libclang-cpp20            libmagickwand-7.q16-10         libtheoradec1      python3.13-gdbm
  gir1.2-gst-plugins-extra-1.0  libclang1-20              liborcus-0.20-0                libtheoraenc1      python3.13-minimal
  gir1.2-totem-1.0              libcrypt-dev              liborcus-parser-0.20-0         libtotem0          software-properties-gtk
  gir1.2-totemplparser-1.0      libdisplay-info2          libpoppler-cpp2                libvpx9            totem
  gnome-power-manager           libfftw3-double3          libpoppler147                  libwmf0.2-7-gtk    totem-common
  gnome-system-monitor          libgdata-common           libprotobuf-c1                 libyelp0           totem-plugins
  grilo-plugins-0.3-base        libgdata22                libproxy1-plugin-gsettings     mecab-ipadic       totem-video-thumbnailer
  imagemagick-7-common          libgpgmepp6t64            libproxy1-plugin-networkmanager mecab-ipadic-utf8 tracker-extract
  libabsl20240722               libgphoto2-l10n           libsframe2                     mecab-utils        ubuntu-wallpapers-questing
Utilice «sudo apt autoremove» para eliminarlos.

Instalando:
  libapache2-mod-php  php  php-mysql

Instalando dependencias:
  libapache2-mod-php8.5  php-common  php8.5  php8.5-cli  php8.5-common  php8.5-mysql  php8.5-readline
```

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ php -v
PHP 8.5.4 (cli) (built: Oct 19 2026 14:35:18) (NTS)
Copyright (c) The PHP Group
Built by Ubuntu
Zend Engine v4.5.4, Copyright (c) Zend Technologies
    with Zend OPcache v8.5.4, Copyright (c), by Zend Technologies
```

- Creación del Virtual Host

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo mkdir /var/www/your_domain
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$
```

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo chown -R $USER:$USER /var/www/your_domain
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$
```

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996: ~ — sudo nano /etc/apache2/sites-available/your_domain.conf
 GNU nano 8.7.1                  /etc/apache2/sites-available/your_domain.conf


                                [ Archivo nuevo ]
^G Ayuda      ^O Guardar    ^F Buscar     ^K Cortar     ^T Ejecutar   ^C Ubicación  M-U Deshacer  M-A Poner marca  M-] A llave
^X Salir      ^R Leer fich. ^\ Reemplazar ^U Pegar      ^J Justificar ^/ Ir a línea M-E Rehacer  M-6 Copiar      ^B Buscar atrás
```

```
  GNU nano 8.7.1                                  /etc/apache2/sites-available/your_domain.conf
<VirtualHost *:80>
        ServerName your_domain
        ServerAlias www.your_domain
        ServerAdmin webmaster@localhost
        DocumentRoot /var/www/your_domain
        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

- Probar el procesamiento de páginas php

```
  GNU nano 8.7.1
<?php
phpinfo();
```

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo rm /var/www/your_domain/info.php
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$
```

- Probar la conexión con la base de datos desde php

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.11-0ubuntu0.26.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

```
mysql> CREATE DATABASE example_database;
Query OK, 1 row affected (0,05 sec)

mysql> CREATE USER 'example_user'@'%' IDENTIFIED BY 'Password123!';
Query OK, 0 rows affected (0,13 sec)

mysql> GRANT ALL ON example_database.* TO 'example_user'@'%';
Query OK, 0 rows affected (0,06 sec)
```

```
mysql> CREATE TABLE example_database.todo_list (
    -> item_id INT AUTO_INCREMENT,
    -> content VARCHAR(255),
    -> PRIMARY KEY(item_id)
    -> );
Query OK, 0 rows affected (0,26 sec)
```

```
mysql> INSERT INTO example_database.todo_list (content) VALUES ("Mi primer elemento de prueba");
Query OK, 1 row affected (0,09 sec)
```

```
  GNU nano 8.7.1                                  /var/www/your_domain/todo_list.php
<?php
$user = "example_user";
$password = "password";
$database = "example_database";
$table = "todo_list";

try {
        $db = new PDO("mysql:host=localhost;dbname=$databate", $user, $password);
        echo "<h2>TODO</h2><ol>";
        foreach($db->query("SELECT content FROM $table") as $row) {
                echo "<li>" . $row['content'] . "</li>";
        }
        echo "</ol>";
} catch (PDOException $e) {
        print "Error!: " . $e->getMessage() . "<br/>!;
        die();
}
```

Un vez concluído debes:

1- Habilitar nuevamente el Virtual host por defecto

sudo a2ensite 000-default

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo a2ensite 000-default
[sudo: authenticate] Contraseña:
Enabling site 000-default.
To activate the new configuration, you need to run:
  systemctl reload apache2
```

2- Deshabilitar al Virtual host “your-domain

sudo a2ensite your_domain

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo a2ensite your_domain
Enabling site your_domain.
To activate the new configuration, you need to run:
  systemctl reload apache2
```

3- Reiniciar Apache

sudo apache2ctl configtest

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo apache2ctl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to s
uppress this message
Syntax OK
```

sudo systemctl reload apache2

```
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$ sudo systemctl reload apache2
rodrigo@rodrigo-Standard-PC-i440FX-PIIX-1996:~$
```

Nota: Para más información sobre apt-get, puedes leer:
http://www.ubuntu-guia.com/2011/01/comando-apt-get-en-ubuntu.html